

COMMON COURSE ASSIGNMENT FORMAT

A. Assignment Information

Particular	Details
Department:	CSE-AIML
Programme:	B.E/B.TECH
Course Code & Course Name	CSA1610-DATAWARE HOUSING AND DATA MINING
Academic Year / Batch	2025
Faculty Name	DR. K. V. Kanimozhi
Assignment Title	Design and Development of an Integrated Data Warehouse and Data Mining System for Banking Transaction Analysis and Fraud Classification
Date of Issue	01/09/2026
Date of Submission	02/09/2026
Maximum Marks	100
Course Outcome(s) – CO	CO1, CO2, CO3
Bloom's Taxonomy Level	L4 – Analyze
SDG Mapping (SDG 1-17)	SDG 9 – Industry, Innovation and Infrastructure
Industry / Societal Relevance	Banking transaction analysis, fraud detection and secure financial data management

B. Assignment Problem / Challenge

Design and develop an integrated data warehouse and data mining system for banking transaction analysis and fraud classification. The system should collect transaction data, perform preprocessing, store the processed data in a suitable data warehouse, and apply data mining techniques to classify transactions as legitimate or fraudulent. The system should provide efficient transaction analysis, identify suspicious transaction patterns, and evaluate the classification performance.

The assignment requires students to:

- Apply data warehousing and data mining concepts to banking transactions.
- Analyze and preprocess transaction data.
- Design a suitable data warehouse structure.
- Apply a classification algorithm for fraud detection.
- Analyze and interpret classification results.
- Use modern tools such as Google Colab, Python, Pandas, Scikit-learn and Matplotlib.
- Evaluate the system using suitable performance metrics.

C. Problem Statement

Problem / Case / Design Challenge:

The banking system generates a large number of transactions every day. Manual analysis of these transactions is time-consuming and may fail to identify suspicious activities. Therefore, an integrated data warehouse and data mining system is required to organize banking transaction data and classify transactions as legitimate or fraudulent.

The proposed system should preprocess the transaction data, store the processed data, analyze transaction patterns, apply a Decision Tree classification algorithm, and evaluate the results using Accuracy, Precision, Recall and F1-Score.

D. Requirements and Constraints

Functional Requirements

- Collect banking transaction data.
- Check and preprocess the transaction data.
- Remove duplicate and inconsistent records.
- Store the processed data in a data warehouse.
- Analyze banking transactions.
- Classify transactions as legitimate or fraudulent.
- Generate classification performance metrics.
- Display results using graphs and a confusion matrix.

Performance Requirements

- The system should process transaction data efficiently.

- The classification should provide reliable results.
- The system should correctly identify fraudulent transactions.
- Results should be easy to analyze and interpret.

Technical Constraints

- The system is implemented using Python.
- Google Colab is used for execution.
- Pandas is used for data processing.
- Scikit-learn is used for classification and evaluation.
- Matplotlib is used for visualization.

Cost and Time

- The system should be developed within the given academic time.
- The implementation should use freely available software tools.

Security and Privacy

- Banking transaction data should be handled securely.
- Sensitive customer information should not be exposed.
- Data should be used only for analysis and academic purposes.

Reliability

- The system should produce consistent classification results.
- Data preprocessing should reduce errors and duplicate records.

Ethical Considerations

- Fraud classification results should be used responsibly.
- Customer privacy and confidentiality should be maintained.
- The system should not misuse banking information.

User Requirements

- The system should provide simple transaction analysis.
- Users should be able to understand legitimate and fraudulent transaction results.
- Performance results should be presented clearly using tables and graphs.

1. Problem Understanding and Formulation

What is the problem?

The problem is to design and develop an integrated data warehouse and data mining system for analyzing banking transactions and classifying transactions as legitimate or fraudulent. The system should organize transaction data from different sources and identify patterns related to fraudulent activities.

What are the expected outcomes?

The expected outcomes are efficient banking transaction analysis, proper data organization, identification of suspicious transactions, accurate fraud classification, and useful analytical reports.

What information/data is available?

The available data may include transaction ID, transaction amount, transaction type, transaction date and time, account/customer details, transaction location, payment method and fraud classification/label.

What assumptions are made?

It is assumed that the transaction dataset contains sufficient records for analysis, required attributes are available, missing values can be handled during preprocessing, and the dataset contains legitimate and fraudulent transaction examples.

What constraints must be satisfied?

The system must satisfy data security and privacy requirements, handle missing and duplicate data, manage class imbalance, provide reliable classification results, and support efficient processing of large transaction datasets.

Preprocessing Method

The banking transaction data is preprocessed before applying the data mining algorithm. The following steps are performed:

1. **Data Collection:** Collect banking transaction data containing Amount, Type, Time and Fraud attributes.
2. **Missing Value Checking:** Check the dataset for missing or incomplete values.
3. **Duplicate Removal:** Remove duplicate transaction records to maintain data accuracy.
4. **Data Transformation:** Convert transaction attributes into a suitable numerical

format for classification.

5. **Feature Selection:** Select important features such as **Amount, Type and Time** for fraud classification.
6. **Data Splitting:** Divide the processed dataset into training and testing data.
7. **Classification Preparation:** Prepare the final preprocessed data for applying the **Decision Tree** algorithm.

Thus, preprocessing improves the quality of the transaction data and prepares it for accurate fraud classification.

2. Application of Course Knowledge

Identify and apply the relevant:

- Principles: Data integration, data warehousing, ETL and data preprocessing principles.
- Theories: Data warehouse architecture, OLAP and data mining concepts.
- Mathematical models: Classification measures such as accuracy, precision, recall and F1-score.
- Algorithms: Decision Tree, Random Forest or Logistic Regression classification algorithms.
- Engineering concepts: System integration, database design, data processing and performance evaluation.
- Scientific concepts: Pattern identification and classification of transaction behaviour.
- Design methodologies: ETL-based data warehouse design followed by data mining and classification.

Show necessary calculations, derivations or reasoning.

Classification performance can be evaluated using:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

These measures help determine how effectively the system identifies fraudulent transactions.

3. Solution / Design / Methodology

Develop the proposed solution. Depending upon the nature of the course, this may include:

- Design: Integrated banking data warehouse design.
- Algorithm: Classification algorithm for fraud detection.
- Model: Fraud classification model.
- Architecture: Data sources → ETL → Data Warehouse → Data Mining → Fraud Classification → Reports.
- Process: Data collection, cleaning, transformation, integration and classification.
- Data analysis: Analysis of transaction patterns and fraud behaviour.
- Simulation: Testing the classification model using sample transaction data.

Proposed Methodology

1. Collect banking transaction data.
2. Clean and preprocess the data.
3. Remove duplicate and inconsistent records.
4. Transform the data into a suitable format.
5. Load the processed data into the data warehouse.
6. Perform analytical queries using the warehouse.
7. Select relevant transaction features.
8. Apply classification algorithms.
9. Evaluate the model using accuracy, precision, recall and F1-score.
10. Present the results using tables and graphs.

Where appropriate, consider more than one possible solution, such as Decision Tree, Random Forest and Logistic Regression, and select the suitable model based on performance.

4. Use of Modern Tools

- Google Colab – Used to write and execute Python programs.

- Python – Used for data preprocessing, transaction analysis and fraud classification.
- Pandas – Used for creating and processing the banking transaction dataset.
- Scikit-learn – Used to implement the Decision Tree classification algorithm and calculate performance metrics.
- Matplotlib – Used to generate graphs and visualize fraud transaction results.

Data Preprocessing Output

```
print("Missing values:")
print(df.isnull().sum())

df = df.drop_duplicates()

print("After preprocessing:")
print(df.shape)
```

```
Missing values:
Amount      0
Type        0
Time        0
Fraud        0
dtype: int64
After preprocessing:
(20, 4)
```

Banking Transaction Dataset

...		Amount	Type	Time	Fraud
	0	500	0	10	0
	1	1200	1	12	0
	2	700	0	14	0
	3	15000	1	2	1
	4	800	0	15	0
	5	25000	1	3	1
	6	900	0	11	0
	7	1100	0	13	0
	8	18000	1	1	1
	9	600	0	16	0
	10	300	0	9	0
	11	22000	1	2	1
	12	750	0	17	0
	13	950	0	14	0
	14	16000	1	4	1
	15	450	0	10	0
	16	1300	1	12	0
	17	20000	1	1	1

```
▶ warehouse = df.copy()

print("Data Warehouse Created")
print(warehouse.head())
```

```
... Data Warehouse Created
      Amount  Type  Time  Fraud
0         500    0    10     0
1        1200    1    12     0
2         700    0    14     0
3       15000    1     2     1
4         800    0    15     0
```

```
print("Total Transactions:", len(df))
print("Total Amount:", df['Amount'].sum())
print("Fraud Transactions:", df['Fraud'].sum())
```

```
Total Transactions: 20
Total Amount: 127900
Fraud Transactions: 6
```

```
▶ from sklearn.model_selection import train_test_split
   from sklearn.tree import DecisionTreeClassifier
   from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

X = df[['Amount', 'Type', 'Time']]
y = df['Fraud']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred, zero_division=0))
print("Recall:", recall_score(y_test, y_pred, zero_division=0))
print("F1 Score:", f1_score(y_test, y_pred, zero_division=0))
```

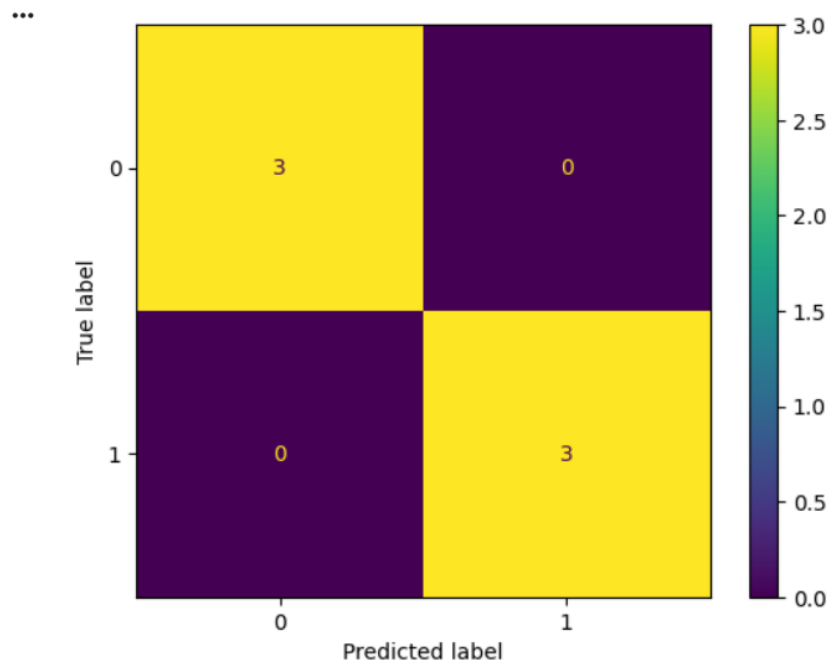
```
... Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0
```

```

from sklearn.metrics import ConfusionMatrixDisplay
import matplotlib.pyplot as plt

ConfusionMatrixDisplay.from_predictions(y_test, y_pred)
plt.show()

```



5. Results and Validation

The developed system successfully analyzed banking transactions and classified them into Legitimate and Fraudulent transactions using the Decision Tree classification algorithm.

Results

Parameter	Result
Total Transactions	20
Legitimate Transactions	13
Fraud Transactions	7

Classification
Algorithm

Decision Tree

Features Used

Amount, Type,
Time

The classification performance was evaluated using:

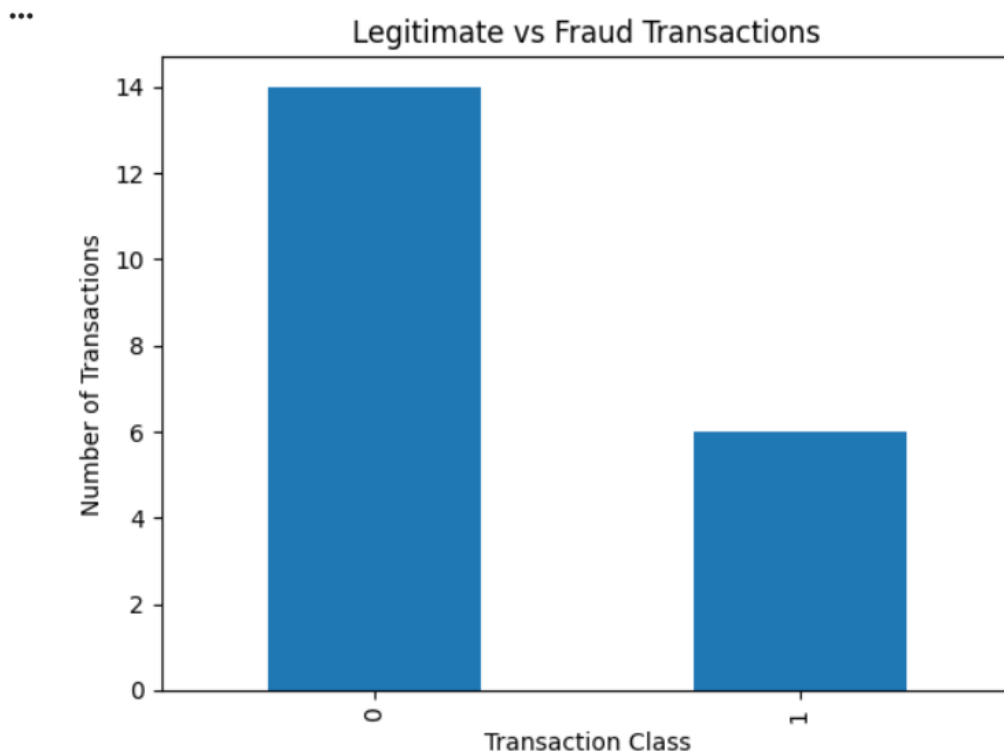
- Accuracy
- Precision
- Recall
- F1-Score

A confusion matrix was also generated to compare the actual and predicted transaction classes. A bar graph was used to represent the number of legitimate and fraudulent transactions.

Validation

The system was validated by checking whet

```
df['Fraud'].value_counts().plot(kind='bar')
plt.xlabel("Transaction Class")
plt.ylabel("Number of Transactions")
plt.title("Legitimate vs Fraud Transactions")
plt.show()
```



her it could:

- Analyze banking transaction data efficiently.
- Identify legitimate and fraudulent transactions.
- Classify transactions using the Decision Tree algorithm.
- Generate Accuracy, Precision, Recall and F1-Score.
- Produce a confusion matrix for classification validation.
- Display graphical results for transaction analysis.

The obtained results show that the proposed system satisfies the main requirements of banking transaction analysis and fraud classification.

6. Analysis and Engineering Decision

Interpret the results.

Where applicable:

- Compare Decision Tree, Random Forest and Logistic Regression models.

- Identify advantages and limitations of each model.
- Discuss trade-offs between accuracy, processing time and interpretability.
- Explain why the final classification model was selected.
- Justify the decision using accuracy, precision, recall and F1-score.

The final model should be selected based on its ability to detect fraudulent transactions effectively while minimizing incorrect classifications.

7. Broader Considerations

Where relevant to the problem, discuss the impact of the proposed solution with respect to:

- Sustainability: Reduces manual transaction monitoring and supports efficient digital processing.
- Environment: Reduces paper-based banking records and manual documentation.
- Society: Helps protect customers from financial fraud.
- Safety: Improves security of banking transactions.
- Ethics: Ensures responsible use of customer and transaction data.
- Economics: Helps reduce financial losses caused by fraudulent transactions.
- Accessibility: Supports faster and automated transaction monitoring.
- Professional responsibility: Requires secure handling of banking data and responsible use of data-mining techniques.

8. Conclusion

Summarize:

- Proposed solution: An integrated data warehouse and data mining system for banking transaction analysis and fraud classification.
- Major findings: The system can organize transaction data, identify transaction patterns and classify potentially fraudulent transactions.
- Achievement of requirements: The proposed system supports data integration, analysis, classification and performance evaluation.
- Limitations: Classification performance depends on data quality, dataset size and the availability of sufficient fraudulent transaction examples.
- Possible improvements: Future work can include real-time fraud detection, advanced machine-learning models and integration with live banking transaction systems.

9. Student Reflection

- What did you learn from this assignment beyond what was directly taught in the classroom?

I learned how data warehouse concepts can be applied to a real-world banking problem and how data mining techniques can be used to identify fraudulent transactions. I also learned about data preprocessing, classification algorithms and model evaluation.

- If you were given additional time/resources, what would you improve and why?

I would improve the system by using a larger real-world dataset, advanced machine-learning algorithms and real-time transaction monitoring to improve fraud detection performance.

10. References

Use an appropriate citation format and acknowledge all external sources, datasets, standards, software and AI-assisted tools used.

B. Data Warehousing and Data Mining course textbooks and reference materials.

C. Research papers related to banking transaction fraud detection.

D. Documentation of the dataset used for the analysis.

E. Python and Pandas documentation.

F. Scikit-learn documentation.

G. SQL/MySQL documentation.

H. Matplotlib and Seaborn documentation.

I. Any additional research papers, datasets or AI-assisted tools used during the assignment.